

Appendix Gaussian Multiple Integrals

We wish first to calculate the multiple integral, over a finite number of real variables ξ_r , of the exponential of a general quadratic function of ξ :

$$\mathcal{J} \equiv \int_{-\infty}^{\infty} \prod_r d\xi_r \exp\{-Q(\xi)\}, \quad (9.A.1)$$

$$Q(\xi) = \frac{1}{2} \sum_{rs} K_{rs} \xi_r \xi_s + \sum_r L_r \xi_r + M, \quad (9.A.2)$$

where K_{rs} , L_r , and M are arbitrary constants, except that the matrix K is required to be symmetric and non-singular. For this purpose, we begin by considering the case where K_{rs} , L_r , and M are all real, with K_{rs} also positive. The result in the general case can then be obtained by analytic continuation.

Any real symmetric matrix can be diagonalized by an orthogonal matrix. Therefore, there is a matrix $\mathcal{S}$ with transpose $\mathcal{S}^T = \mathcal{S}^{-1}$ such that

$$(\mathcal{S}^T K \mathcal{S})_{rs} = \delta_{rs} \kappa_r. \quad (9.A.3)$$

Because K is assumed positive and non-singular, the eigenvalues κ_r are positive-definite. We can use the matrix $\mathcal{S}$ to perform a change of variables:

$$\xi_r = \sum_s \mathcal{S}_{rs} \xi'_s. \quad (9.A.4)$$

The Jacobian $|\text{Det } \mathcal{S}|$ of this transformation is unity, so the multiple integral (9.A.1) is now given by a product of ordinary integrals:

$$\begin{aligned} \mathcal{J} &= e^{-M} \prod_r \int_{-\infty}^{\infty} d\xi'_r \exp \left\{ -\frac{\kappa_r}{2} \xi'^2_r - (\mathcal{S}^T L)_r \xi'_r \right\} \\ &= e^{-M} \prod_r \sqrt{\frac{2\pi}{\kappa_r}} \exp \left\{ \frac{1}{2\kappa_r} (\mathcal{S}^T L)_r^2 \right\}. \end{aligned} \quad (9.A.5)$$

But the determinant and the reciprocal of Eq. (9.A.3) give

$$\text{Det } K = \prod_r \kappa_r, \quad K_{rs}^{-1} = \sum_t \mathcal{S}_{rt} \mathcal{S}_{st} \kappa_t^{-1},$$

so Eq. (9.A.5) may be written

$$\mathcal{J} = \left(\text{Det } \frac{K}{2\pi} \right)^{-1/2} \exp \left\{ \frac{1}{2} \sum_{rs} L_r L_s K_{rs}^{-1} - M \right\}. \quad (9.A.6)$$

Eq. (9.A.1) defines a function of K_{rs} , L_r , and M that is analytic in K_{rs} in a finite region around the surface where K_{rs} is real and positive, where the integral converges, and for such K_{rs} is analytic everywhere in L_r and M . Since (9.A.6) equals (9.A.1) for K_{rs} , L_r , and M all real, with K_{rs} also positive, Eq. (9.A.6) provides an analytic continuation of Eq. (9.A.1) to the whole complex plane, with a cut required by the square root. The sign of the

square root is fixed by this analytic continuation. In field theory K_{rs} is actually imaginary, except for a small real part due to the ‘ $i\epsilon$ term’.

It is useful to express Eq. (9.A.6) in terms of the stationary point of the function (9.A.2):

$$\bar{\xi}_r = - \sum_s K_{rs}^{-1} L_s, \quad (9.A.7)$$

$$\frac{\partial Q(\xi)}{\partial \xi_r} = 0 \quad \text{at} \quad \xi = \bar{\xi}, \quad (9.A.8)$$

as

$$\mathcal{J} = \left(\text{Det} \frac{K}{2\pi} \right)^{-1/2} \exp\{-Q(\bar{\xi})\}. \quad (9.A.9)$$

This is the result to remember: *Gaussian integrals can be evaluated up to a determinant factor by setting the integration variable equal to the point where the argument of the exponential is stationary.*

We next wish to use this result to calculate the integrals

$$I_{r_1 r_2 \dots r_{2N}} \equiv \int \left(\prod_r d\xi_r \right) \xi_{r_1} \xi_{r_2} \dots \xi_{r_{2N}} \exp \left\{ -\frac{1}{2} \sum_{rs} K_{rs} \xi_r \xi_s \right\}. \quad (9.A.10)$$

(Integrals of this sort with an odd number of ξ -factors in the integrand obviously vanish.) From the power-series expression of $\exp(-\sum_r L_r \xi_r)$ in Eq. (9.A.1), we have the sum rule

$$\begin{aligned} & \sum_{N=0}^{\infty} \sum_{r_1 r_2 \dots r_{2N}} \frac{1}{(2N)!} I_{r_1 r_2 \dots r_{2N}} L_{r_1} L_{r_2} \dots L_{r_{2N}} \\ &= \int \left(\prod_r d\xi_r \right) \exp \left\{ -\sum_r L_r \xi_r - \frac{1}{2} \sum_{rs} K_{rs} \xi_r \xi_s \right\} \\ &= \left[\text{Det} \left(\frac{K}{2\pi} \right) \right]^{-1/2} \exp \left\{ \frac{1}{2} \sum_{rs} L_r L_s K_{rs}^{-1} \right\} \\ &= \left[\text{Det} \left(\frac{K}{2\pi} \right) \right]^{-1/2} \sum_{N=0}^{\infty} \frac{1}{N! 2^N} \left(\sum_{rs} L_r L_s K_{rs}^{-1} \right)^N. \end{aligned} \quad (9.A.11)$$

Comparing the coefficients of $L_{r_1} L_{r_2} \dots L_{r_{2N}}$ on both sides, we see that $I_{r_1 r_2 \dots r_{2N}}$ must be proportional to a sum of products of elements of K^{-1} , which symmetry requires to take the form

$$I_{r_1 r_2 \dots r_{2N}} = c_N \sum_{\substack{\text{pairings} \\ \text{of } r_1 \dots r_{2N}}} \prod_{\text{pairs}} (K^{-1})_{\text{paired indices}}. \quad (9.A.12)$$

Here the sum is over all ways of pairing the indices $r_1 \dots r_{2N}$, with two pairings being considered the same if they differ only by the order of the pairs, or by the order of indices within a pair. To calculate the constant factor c_N , we note that the number ν_N of terms in the sum over pairings in Eq. (9.A.12) is equal to the number $(2N)!$ of permutations of the

indices, divided by the number of ways $N!$ of permuting index pairs and by the number 2^N of permutations within index pairs

$$\nu_N = \frac{(2N)!}{N!2^N}. \quad (9.A.13)$$

Therefore, Eq. (9.A.12) gives

$$\sum_{r_1 r_2 \dots r_{2N}} L_{r_1} L_{r_2} \dots L_{r_{2N}} I_{r_1 r_2 \dots r_{2N}} = \nu_N c_N \left(\sum_{rs} L_r L_s K_{rs}^{-1} \right)^N. \quad (9.A.14)$$

Comparing this with Eq. (9.A.11) shows that the factors $(2N)!$ and $N!2^N$ are cancelled by ν_N , leaving us with

$$c_N = \left[\text{Det} \left(\frac{K}{2\pi} \right) \right]^{-1/2}. \quad (9.A.15)$$

For instance,

$$I_{r_1 r_2} = I_0 (K^{-1})_{r_1 r_2}, \quad (9.A.16)$$

$$\begin{aligned} I_{r_1 r_2 r_3 r_4} = I_0 [& (K^{-1})_{r_1 r_2} (K^{-1})_{r_3 r_4} + (K^{-1})_{r_1 r_3} (K^{-1})_{r_2 r_4} \\ & + (K^{-1})_{r_1 r_4} (K^{-1})_{r_2 r_3}], \end{aligned} \quad (9.A.17)$$

and so on, where I_0 is the integral with no indices

$$\begin{aligned} I_0 &= \int \left(\prod_r d\xi_r \right) \exp \left\{ -\frac{1}{2} \sum_{rs} K_{rs} \xi_r \xi_s \right\} \\ &= \left[\text{Det} \left(\frac{K}{2\pi} \right) \right]^{-1/2}. \end{aligned} \quad (9.A.18)$$

Problems

1. Consider a non-relativistic particle of mass m , moving along the x -axis in a potential $V(x) = m\omega^2 x^2/2$. Use path-integral methods to find the probability that if the particle is at x_1 at time t_1 , then it is between x and $x + dx$ at time t .
2. Find the wave function in field space of a state consisting of a single spinless particle of mass $m \neq 0$. Use the result to derive the Feynman rules for emission or absorption of such a particle.
3. Find the wave function in field space of the vacuum in the theory of a neutral vector field of mass $m \neq 0$. Use the result to derive the form of the $i\epsilon$ terms in the propagator of this field.
4. The Lagrangian density of the free spin 3/2 Rarita–Schwinger field ψ^μ is

$$\mathcal{L} = -\bar{\psi}^\mu (\gamma^\nu \partial_\nu + m) \psi_\mu - \frac{1}{3} \bar{\psi}^\mu (\gamma_\mu \partial_\nu + \gamma_\nu \partial_\mu) \psi^\nu + \frac{1}{3} \bar{\psi}^\mu \gamma_\mu (\gamma^\sigma \partial_\sigma - m) \gamma^\nu \psi_\nu.$$

Use path-integral methods to find the propagator of this field.

References

1. R. P. Feynman, *The Principle of Least Action in Quantum Mechanics* (Princeton University, 1942; University Microfilms Publication No. 2948, Ann Arbor). Also see R. P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals* (McGraw-Hill, New York, 1965). For a general reference, see J. Glimm and A. Jaffe, *Quantum Physics—A Functional Integral Point of View*, 2nd edn (Springer-Verlag, New York, 1987).
2. P. A. M. Dirac, *Phys. Zeits. Sowjetunion* **3**, 62 (1933).
3. R. P. Feynman, *Rev. Mod. Phys.* **20**, 367 (1948); *Phys. Rev.* **74**, 939, 1430 (1948); **76**, 749, 769 (1949); **80**, 440 (1950).
4. L. D. Faddeev and V. N. Popov, *Phys. Lett.* **B25**, 29 (1967). Also see R. P. Feynman, *Acta Phys. Pol.* **24**, 697 (1963); S. Mandelstam, *Phys. Rev.* **175**, 1580, 1604 (1968).
5. B. De Witt, *Phys. Rev. Lett.* **12**, 742 (1964).
6. G. 't Hooft, *Nucl. Phys.* **B35**, 167 (1971).
7. I. S. Gerstein, R. Jackiw, B. W. Lee, and S. Weinberg, *Phys. Rev.* **D3**, 2486 (1971).
8. L. D. Faddeev, *Teor. Mat. Fizika* **1**, 3 (1969); translation in *Theor. Math. Phys.* **1**, 1 (1970).
- 8a. J. Schwinger, *Proc. Nat. Acad. Sci.* **44**, 956 (1958).
- 8b. K. Osterwalder and R. Schrader, *Phys. Rev. Lett.* **29**, 1423 (1972); *Commun. Math. Phys.* **31**, 83 (1973); *Commun. Math. Phys.* **42**, 281 (1975). The Osterwalder–Schrader axioms require smoothness, Euclidean covariance, ‘reflection positivity,’ permutation symmetry, and cluster decomposition.
9. This section was largely inspired by discussions with J. Polchinski.
10. F. A. Berezin, *The Method of Second Quantization* (Academic Press, New York, 1966).
11. M. G. G. Laidlaw and C. M. De Witt, *Phys. Rev.* **D3**, 1375 (1970).
12. J. M. Leinaas and J. Myrheim, *Nuovo Cimento* **37B**, 1 (1977).
13. Y. Ohnuki and S. Kamefuchi, *Phys. Rev.* **170**, 1279 (1968); *Ann. Phys.* **51**, 337 (1969); K. Drühl, R. Haag, and J. E. Roberts, *Commun. Math. Phys.* **18**, 204 (1970).
14. The braid group was introduced by E. Artin. See *The Collected Papers of E. Artin*, ed. by S. Lang and J. E. Tate (Addison-Wesley, Reading, MA, 1965).
15. F. Wilczek, *Phys. Rev. Lett.* **49**, 957 (1982); K. Fredenhagen, M. R. Gaberdiel, and S. M. Rüger, Cambridge preprint DAMTP-94-90 (1994). Also see J. M. Leinaas and J. Myrheim, Ref. 12.